Ultrafast Electron Dynamics – Concept

- Attosecond (10^{-18} s) ⇒ Timescale of pure electron motion
- Electronic motion

  ⇒ Quantum description of electrons

  ⇒ Mechanisms underlying **all** chemical processes

  ⇒ Spatial and temporal evolution of electrons

Ultrafast Electron Dynamics – Key Quantity

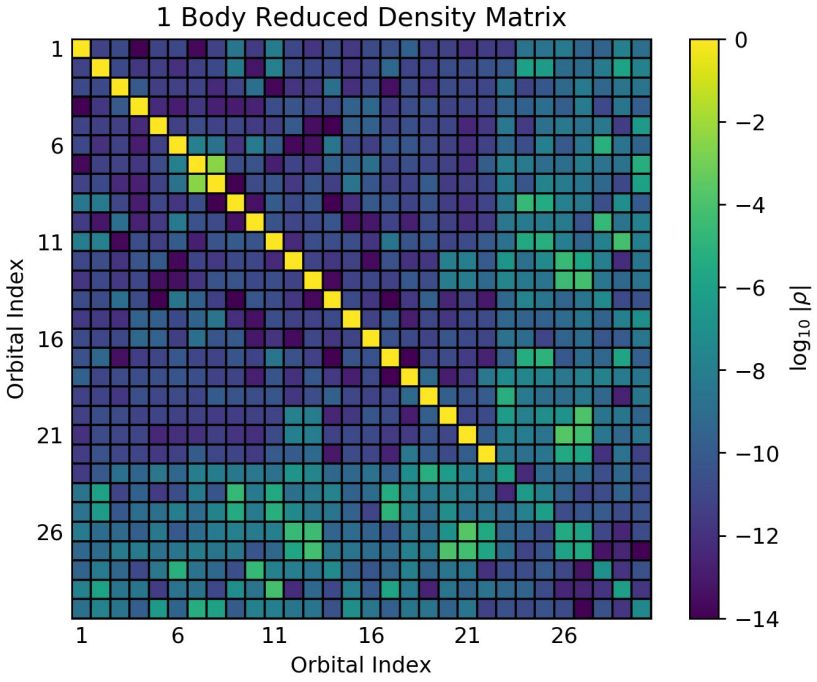
Tracking individual electrons is complicated so we work with a matrix quantity **D**(t) which encodes the **composite all-electronic state** of the system at a given time.

$$\mathbf{D}(t) \in \mathbb{C}^{N\times N}$$

Electron density is a fundamental quantity:

- Function of 3D real space (and time) ⇒ asymptotic decay, cusps at nuclei ⇒ exponential decay
- All physical observable quantities eg, dipole moment, can be extracted from this single quantity
- Diagonal elements ⇒ Population

  Off-diagonal elements ⇒ Correlations and coherence



Ultrafast Electron Dynamics – Key Equation

- Liouville–von Neumann equation of motion:

$$i\frac{d}{dt}\mathbf{D}(t) = \left[\mathbf{F}(\mathbf{D}(t),t),\ \mathbf{D}(t)\right]$$

commutator

- Fock matrix **F** depends on **D** (and other molecule and light dependent terms)

$$\mathbf{F}(t) = \mathbf{H}^{\mathrm{core}} + \mathbf{J}[\mathbf{D}(t)] + \mathbf{V}^{\mathrm{xc}}[\mathbf{D}(t)] + \mathbf{V}^{\mathrm{ext}}(t)$$

- Time propagation: Second-order Magnus propagator

$$\mathbf{D}(t+\Delta t) = \mathbf{U}\,\mathbf{D}(t)\,\mathbf{U}^{\dagger}$$

$$\mathbf{U} = \exp\left[-i\,\Delta t\,\mathbf{S}^{-1}\mathbf{F}\left(t+\tfrac{\Delta t}{2}\right)\right]$$

- Numerically, at every time-step, we: build **F**, construct **U**, update **D**

- Total number of timesteps: ~ 10^4 – 10^5
  Matrix dimension: N x N (N ~ 10^2 – 10^3)
  Each time-step **Fock F** construction is atleast $\mathcal{O}(N^3)$

→ Multi-day HPC calculations

Where AI/ML can help

1. Code Acceleration

Handling the computational bottle-necks:

- Fock construction
- Density Propagation

2. Dimensionality Reduction

Unsupervised learning for analysis of the dynamics

- Principal Component Analysis
- More advanced AI/ML approaches